

Statistical Modeling and Inferencing

Assignment-1

TOTAL MARKS: 20**DUE DATE: 8th August 2026, 11:59 PM****Assignment Weightage: 20% of Final Grade**

Submission Format

Submission: Submit a Jupyter Notebook (.ipynb) file containing your complete analysis. If using Google Colab, either download and submit the .ipynb file, or share the Colab link with “Anyone with the link can view” access enabled. Use markdown cells to document your findings, interpretations, and insights throughout the notebook.

File Naming: YourBITSID_SMI_Assignment.ipynb

Dataset Selection

Choose **one** of the following datasets for your analysis:

Dataset 1: Diamonds Dataset

- **URL:** [diamonds.csv](#) (~53,940 observations)
- **Features:** carat, cut, color, clarity, depth, table, x, y, z → predict price

Dataset 2: Mall Customer Segmentation Dataset

- **URL:** [Mall_Customers.csv](#) (~200 observations)
- **Features:** CustomerID, Gender, Age, Annual Income (k\$), Spending Score (1–100)

Assignment Components

Part 1: Data Exploration and Preparation (6 marks)

Load your dataset and present descriptive statistics with appropriate visualizations exploring distributions and relationships. Conduct a correlation analysis to understand how variables relate to each other and your target variable. Identify any data quality issues such as missing values and outliers, documenting your decisions on how to handle them. Encode categorical variables if present and describe any feature transformations, providing justification for your preprocessing choices.

Part 2: Model Development and Validation (10 marks)

Apply appropriate analytical techniques based on your chosen dataset and validate your approach systematically.

For Dataset 1: Investigate relationships through regression modeling. Examine correlations among predictors and consider dimensionality reduction approaches if needed. Develop multiple models with different predictor combinations or transformations. Compare models using appropriate

statistical metrics. Address multicollinearity concerns and identify influential features. Select your best model with clear justification.

For Dataset 2: Perform clustering analysis to identify natural customer groupings. Determine the optimal number of clusters using methods such as the Elbow method or Silhouette analysis. Apply both K-Means and Hierarchical clustering, and visualize patterns using dimensionality reduction methods and dendrograms where appropriate. Evaluate the quality of your chosen clustering solution. Document your analytical decisions and present key outputs clearly.

Part 3: Interpretation and Insights

(4 marks)

Interpret your key findings clearly and provide actionable insights. What patterns emerged from your analysis? Discuss the practical implications of your results, acknowledge limitations of your analysis, and provide recommendations based on your findings. Consider what additional data or analysis might strengthen your conclusions.

Document Organization

Your submission should follow this structure:

1. **Title Page:** Name, roll number, chosen dataset, and submission date
2. **Colab Link (if applicable):** If submitting via Google Colab, include the shared link at the top
3. **Summary:** Brief overview of your approach and key findings (2-3 paragraphs)
4. **Part 1:** Data exploration and preparation with visualizations and documentation
5. **Part 2:** Model development, comparison, validation, and diagnostics
6. **Part 3:** Interpretation, insights, limitations, and recommendations

Timeline

Assignment Released: 12th July 2026

Submission Deadline: 8th August 2026, 11:59 PM

Questions? Post on the course discussion forum.